

F-65 Megawing

Audio, Sound Effects and Music Engineering White Paper v1.0

Document Type: Technical / Product Audio Engineering White Paper

Platform: MEGA65 – 45GS02 + VIC-IV audio subsystem

Audience: Product/audio designers, engine implementers, C developers, 45GS02 platform engineers, asset-tool authors

Status: FINAL – HUMAN-REVIEWED v1.0

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Nature: Detailed audio-design baseline for F-65 Megawing implementation, integration, and specification maintenance.

Approval effect: Version 1.0 is the human-reviewed audio engineering baseline. It remains subordinate to controlling program authority and configuration-control rules.

Authority and precedence

This paper is subordinate to the controlling F-65 architecture and approved specification set. It cannot relax architecture, timing, memory, ownership, deterministic-simulation, DMA, resource-staging, sensor, graphics, or platform requirements established at higher authority.

1. Purpose

Audio is a primary presentation system for F-65 Megawing.

The product depends on audio to communicate:

- aircraft energy and engine state;
- airspeed and aerodynamic loading;
- system operation and failure;
- radar and missile threat state;
- weapon state;
- carrier and runway interaction;
- RIO, AIC, and ATC information;
- combat impact and scale; and
- the retro-synthwave identity of the game.

The intended result is not a generic collection of game sound effects layered beneath graphics. Audio is a protected cockpit-information and immersion service operating alongside HUD and critical-warning presentation.

The player should be able to infer meaningful aircraft and tactical state by ear while still experiencing the deliberate limited-polyphony character of late-1980s and early-1990s computer audio.

2. Architecture Boundary

2.1 Authoritative simulation versus presentation

Audio is presentation.

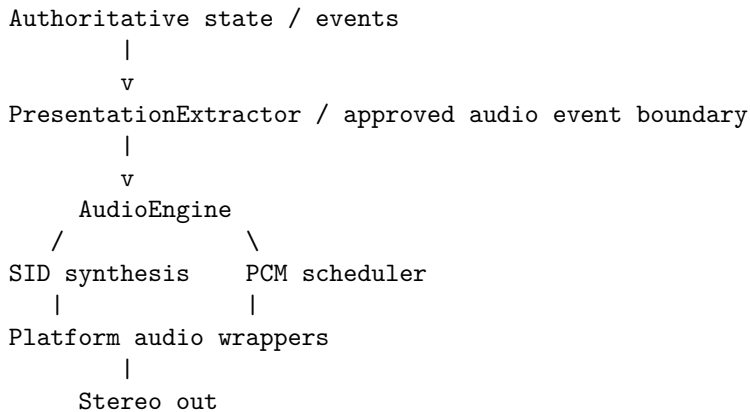
Audio playback, missing samples, music state, SID voice stealing, PCM preemption, panning, ducking, dropped effects, or audio-device timing shall never change:

- flight physics;
- weapons;
- sensors;
- track state;
- AI or RIO decisions;
- damage;
- mission state;
- scoring;
- RNG state; or
- authoritative checksums.

Simulation and subsystem state produce semantic audio events and bounded continuous parameters.

AudioEngine converts those inputs into SID and PCM presentation.

Conceptually:



AudioEngine does not reach into private FlightDynamics, SensorAndTrack, WeaponAndDamage, AI, or systems arrays.

2.2 Hardware ownership

Only the protected platform/audio layer writes SID registers or audio-DMA hardware registers.

Gameplay and simulation modules request sound by semantic event or parameter.

Examples:

- ENGINE_AUDIO_STATE
- RWR_SEARCH
- RWR_TRACK
- RWR_LAUNCH
- ACTIVE_SEEKER_WARNING
- MASTER_CAUTION
- STALL_WARNING
- GEAR_TRANSIT
- TOUCHDOWN
- CATAPULT_LAUNCH

- WEAPON_RELEASE
- GUN_BURST
- HEATER_ACQUIRE
- RIO_CALLOUT
- ATC_CALLOUT

The hardware layer contains no gameplay decision logic.

3. MEGA65 Audio Resource Model

The target platform provides two complementary sound mechanisms:

1. Four SID synthesizers, providing twelve SID oscillator voices.
2. Four hardware PCM/audio-DMA playback channels feeding the digital stereo path.

The architectural intent is:

SID - continuous - procedural - repeating - parameter-driven - warning tones - musical synthesis

PCM - speech - characteristic mechanical transients - characteristic environmental transients - selected combat impacts - sounds whose spectral content is inefficient to synthesize

SID should be used whenever it can convincingly express the sound.

PCM is a scarce enhancement resource rather than the default sound generator.

4. SID Voice Baseline

4.1 Logical chip allocation

The normal allocation is:

SID	Primary role	Voices
SID 1	Music left / primary musical synthesis	3
SID 2	Music right / complementary musical synthesis	3
SID 3	Continuous aircraft / environment	3
SID 4	Warnings, RWR, procedural cockpit and transient effects	3

This is the preferred ownership model rather than an absolute hardware partition.

Priority arbitration may temporarily borrow or silence a lower-priority voice.

No gameplay subsystem directly owns a physical SID voice.

4.2 Why music receives two SIDs

F-65's music is part of the product identity.

Two SIDs permit:

- stereo placement;
- independent complementary timbres;
- doubled or answering parts;
- wider lead and rhythm treatment;
- independent filter behavior between the two musical chips; and
- graceful degradation from six available music voices toward the protected three-voice musical core.

The minimum recognizable music identity is three simultaneous logical parts:

1. percussion;
2. bass/rhythm;
3. lead.

Additional voices provide doubling, harmony, arpeggiation, pad, echo-like responses, or stereo reinforcement.

4.3 Continuous-flight SID

SID 3 normally carries:

- engine-body / low-frequency propulsion layer;
- higher-frequency engine spool layer; and
- wind / slipstream layer.

The engine synthesizer consumes actual engine operating state rather than raw throttle position.

Relevant parameters include, as available from the approved snapshot/interface:

- Engine 1 RPM/N1 state;
- Engine 2 RPM/N1 state;
- military/afterburner state;
- engine failure or flameout state;
- airspeed;
- AoA or buffet state where appropriate; and
- cockpit/external-view mix selection.

Twin-engine asymmetry must therefore be audible without requiring a separate PCM recording for every engine condition.

A practical implementation may collapse both engines into a bounded two-layer procedural model while retaining deterministic response to asymmetric RPM or engine-out state.

4.4 Warning/effect SID

SID 4 normally carries:

- master caution;
- critical warning tones;
- RWR search/track/launch families;
- active-radar-seeker warning;
- heater seeker/acquisition tone;
- stall or ground-impact warning;
- gear/flap/hook procedural cues;
- cannon pulse train when PCM is unavailable or preempted; and
- other cockpit tones.

Simultaneous tones are prioritized rather than allowed to create uncontrolled voice contention.

5. PCM / Audio-DMA Model

5.1 Four-channel pooled scheduler

The four PCM channels form one scheduler-controlled pool.

They are not permanently assigned as two fixed stereo pairs.

Most F-65 PCM assets should be mono.

Stereo playback is permitted only where the perceptual value justifies consuming two channels.

Possible stereo use includes exceptional major environmental or launch events. Speech, cockpit mechanisms, weapon transients, and most aircraft effects remain mono unless evidence shows otherwise.

The scheduler chooses the available physical channel and output routing according to:

- priority;
- current channel state;
- retrigger rule;
- pan requirement;
- safe cut point;
- sample residency; and
- hardware contention limits.

5.2 Preferred PCM content

PCM is preferred for sounds such as:

- catapult launch transient;
- arrestment / deck impact where required;
- touchdown and tire chirp/screech;
- heavy missile-release or rocket-motor ignition transient;
- selected cannon report layer;
- landing-gear mechanical clunk;
- hook or chute mechanical event;
- crash / heavy impact;
- short RIO phrases;
- short ATC/AIC phrases; and
- any similarly short event that cannot be rendered convincingly by SID synthesis.

A PCM asset never becomes mandatory merely because it has been authored.

Every gameplay-essential event has an approved SID, text, or protected visual fallback.

6. Priority and Preemption

The existing protected service hierarchy remains controlling.

Priority	Content	Audio behavior
P0	Missile warning, fire, critical engine/FCS state, stall/ground-impact warning	Immediate protected service; may preempt any lower class
P1	Other immediate aircraft-safety cautions	May preempt ordinary speech/effects
P2	Critical RIO defensive, Bingo/fuel, recovery callouts	PCM when available; immediate text/tone fallback
P3	Ordinary RIO, ATC, AIC and mission speech	Queue, abbreviate, preempt, or fall back to text
P4	Weapon transients, impacts, ambience, decorative effects	First presentation audio to drop

Music participates in resource arbitration as a persistent background service but never outranks P0 or P1 safety presentation.

6.1 Critical-warning rule

A P0 warning may:

- steal a SID voice;
- stop or duck music;
- immediately terminate lower-priority PCM;
- defer a weapon transient; or
- interrupt speech.

P0 latency cannot be sacrificed to preserve musical continuity.

6.2 Music degradation

Under ordinary load:

6 music voices available

↓

optional music voices stolen

↓

3-voice musical core preserved

↓

temporary duck / voice interruption

↓

temporary silence only if required by protected warning service

After contention clears, musical voices are restored at a musically valid boundary.

The music sequencer must recover without stuck notes, permanently lost parts, or corrupted envelope/filter state.

6.3 Speech interruption

Speech may normally be interrupted only at an authored safe cut point.

P0 may terminate speech immediately.

The corresponding text remains available regardless of whether the spoken sample:

- starts;
- completes;
- is preempted;
- is missing; or
- is dropped.

7. Cockpit and Flight Sound Design

7.1 Engines

Engine sound is procedural by default.

Desired perceptual behavior:

- idle has a low mechanical/airflow body;
- RPM rise increases pitch, brightness, and modulation rate;
- military power produces a stronger high-frequency layer;
- afterburner materially changes timbre and intensity;
- spool-up and spool-down lag actual engine state rather than instantly following the throttle;
- engine-out changes the resulting sound immediately according to actual engine behavior;
- asymmetric engine state remains audible; and
- shutdown removes propulsion layers according to actual spool state rather than a canned timer.

Exact oscillator types, pulse width, filter cutoff, resonance, gain curves, and update quantization are authored audio parameters.

7.2 Wind and aerodynamic loading

Wind uses filtered noise.

Its gain and spectral character may depend on:

- indicated or true airspeed as appropriate to the audio model;
- configuration;
- AoA;
- sideslip;
- canopy/view presentation; and
- buffet state.

The objective is not atmospheric CFD audio.

The objective is an immediately perceptible relationship between aircraft energy and sound.

7.3 Buffet / high-AoA cue

Where the approved flight model exposes buffet state, AudioEngine may add:

- low-frequency amplitude modulation;
- filtered noise modulation;
- intermittent structural/procedural vibration cue.

The cue represents actual model state.

It cannot predict or create stall state independently.

8. Warning and Sensor Audio

8.1 Master caution

Master caution uses a distinctive short two-tone or gated SID pattern.

Generic master caution is normally P1.

A specific underlying P0 condition retains P0 behavior regardless of whether the master-caution annunciation itself is generic.

8.2 RWR

The audio vocabulary must distinguish at minimum:

- search / surveillance;
- track or fire-control attention;
- launch / missile support warning; and
- active radar seeker warning where SensorAndTrack reports a valid received emission.

Exact emitter-specific musical pitches are not required.

The priority is recognition under load.

Launch and valid active-seeker warnings receive the strongest protection.

Search traffic must not continuously consume warning resources when tactically irrelevant.

8.3 IR seeker tone

The heater seeker tone represents the player's own passive IR weapon seeker.

It may communicate:

- no valid acquisition;
- weak candidate;
- acquisition/growl;
- valid firing-quality acquisition where that concept is exposed.

It does not indicate an incoming IR missile.

The F-65 has no dedicated IR missile-approach-warning system in the current design.

9. Weapon and Carrier Effects

9.1 Catapult

Catapult launch should be one of the strongest non-warning PCM events in the game.

Preferred construction:

PCM launch transient
+
short SID mechanical/noise tail
+
existing engine/wind state

It remains presentation-only and may be dropped or simplified under protected contention.

9.2 Missile release

A missile launch may combine:

- short PCM release/ignition transient; and
- SID noise or filtered tail.

The sound originates from an actual release event.

A rejected release request produces no launch sound.

9.3 Cannon

The cannon may be represented by:

- SID pulse/noise train synchronized to the approved grouped-fire presentation; or
- a short PCM burst layered with SID.

Audio does not instantiate individual projectile physics.

The weapon model remains authoritative.

9.4 Landing gear and hook

Gear motion may combine procedural hydraulic/noise texture with short mechanical end-stop PCM.

Separate events should be available for:

- gear command;
- gear transit if perceptually useful;
- gear lock;
- hook movement; and

- exceptional damaged or incomplete state if approved.

The sound follows actual actuator/system state rather than merely the player's switch input.

9.5 Touchdown and arrestment

Touchdown may use a short PCM tire/deck transient.

Carrier arrestment may add a distinct arresting-load presentation cue derived from the actual contact/arrestment event.

10. Music Architecture

10.1 Product character

Music is retro synthwave with a deliberate SID identity.

It should sound like music written for the hardware rather than modern music reduced until it fits.

The core instrument architecture is:

Percussion

One primary logical voice.

Kick, snare, tom/impact and hat-like articulation are created through:

- noise and/or pulse;
- pitch changes;
- envelope changes; and
- brief gated events.

Bass / rhythm

One primary logical voice.

Preferred character:

- pulse or square;
- controlled PWM;
- filter movement;
- strong low-mid rhythmic foundation.

Lead

One primary logical voice.

Preferred character:

- saw;
- triangle/saw variants;
- vibrato;
- arpeggio or melodic hook;
- optional doubled or answering stereo voice.

10.2 Optional musical voices

When resources permit, additional voices may provide:

- pad/atmosphere;
- harmony;
- octave doubling;
- arpeggiation;

- stereo answering phrase;
- pseudo-delay; or
- additional percussion articulation.

These voices are the first music voices surrendered during contention.

10.3 Track roles

The soundtrack baseline contains at least six functional musical roles:

1. Menu / briefing.
2. Sortie start / launch / high-energy.
3. Cruise / nostalgic.
4. Threat / dark tactical.
5. Carrier approach / recovery.
6. Victory / debrief.

Additional tracks and variations may be authored later.

All tracks are written against the same constrained instrument architecture so soundtrack identity remains coherent.

10.4 Music state machine

Music changes by presentation state/event rather than directly by arbitrary game-module calls.

Transitions should occur at musically meaningful boundaries where practical.

The music system supports:

- start;
- stop;
- loop;
- section change;
- track change;
- duck;
- temporary voice reduction;
- restore; and
- pause/resume.

A change in music can never change simulation timing or state.

11. Speech, RIO, AIC and ATC

Speech is additive to the authoritative text system.

Text remains the information authority.

The initial strategy is a bounded vocabulary of short reusable phrases rather than unrestricted recorded dialogue.

Suitable PCM phrase classes include:

- warning nouns;
- bearings/headings;
- short tactical verbs;
- weapon calls;
- Bingo/Joker;
- recovery cues;
- launch/recommit/defend calls;
- short ATC acknowledgements.

Long natural-language dialogue is not assumed to fit the resident audio budget.

Exact phrase vocabulary and recording scope remain a later product/audio decision constrained by measured cache, storage, staging and latency evidence.

Critical calls must remain intelligible when speech is absent.

12. PCM Format and Memory

12.1 Audio cache

The current pre-measured audio sample-cache planning allocation is 12 KB inside the shared resource-staging region.

This is a ceiling to prove against, not permission to assume every desired sample can remain resident.

The final allocation remains subject to simultaneous graphics/resource/audio/DMA measurement.

12.2 Asset declaration

Every PCM asset declares at minimum:

- asset_id
- semantic_audio_class
- priority
- encoding
- signedness
- bit_depth
- sample_rate
- sample_count
- decoded_bytes
- alignment
- loop_start / loop_end if applicable
- safe_cut_points if speech
- default_gain
- pan/routing policy
- retrigger policy
- resident/preload/staged class
- fallback

Exact binary field sizes are generated later.

12.3 Formats

The platform can support multiple PCM representations.

The white paper does not freeze:

- 4-bit versus 8-bit;
- signed versus unsigned;
- sample rate;
- maximum sample duration;
- stereo use;
- loop granularity; or
- aggregate active PCM bandwidth.

These close through measurement and product/audio review.

Short low-rate PCM should be preferred when its perceptual quality is sufficient.

13. Residency and Staging

Audio hardware never consumes an Attic address directly.

The resource flow is:

```
Mission package / cold asset
      |
      v
Attic RAM
      |
ResourceManager bounded staging
      |
      v
chip-RAM audio cache
      |
      v
AudioEngine / platform audio
```

Tactical disk reads are prohibited.

Attic-to-chip staging is permitted only through the existing resource/DMA ownership model.

An audio event must not stall simulation waiting for an optional sample.

If the desired sample is unavailable:

- use the SID fallback;
- use text;
- use a simpler resident sample; or
- omit the presentation-only transient.

Warnings cannot depend on optional sample residency.

14. Update Cadence

The authoritative simulation remains exactly 100 Hz.

Audio service is an independent protected service.

Continuous source parameters may be refreshed less frequently than the 100 Hz physics tick if measurement shows that doing so is perceptually transparent.

Examples include:

- engine synthesis parameters;
- wind filter state;
- music envelopes/sequencer;
- low-priority ambience.

Critical warning activation follows the protected latency contract and is not delayed to a convenience update cadence.

Exact:

- service period;
- engine-parameter refresh;
- music tick rate;
- PCM service rate;

- IRQ/DMA interaction; and
- maximum warning latency

remain measured values.

15. Views and Spatial Presentation

F-65 does not require a general 3D positional-audio engine.

Stereo is used deliberately and cheaply.

Permitted presentation effects include:

- mild left/right engine asymmetry;
- selected threat bearing emphasis;
- cockpit versus chase gain/EQ differences;
- stereo music placement;
- stereo environmental width.

Camera or view state may change presentation mix only.

It cannot change which event physically occurred.

The cockpit mix remains the primary authored experience.

16. Full Pause

At full pause:

- authoritative simulation audio events stop advancing;
- active physical-event loops may be silenced or held according to the approved presentation rule;
- no new simulation-driven warning or weapon event is invented;
- pause/menu music may continue or transition;
- UI sounds remain presentation-only.

On resume, stale one-shot physical events are not replayed unless explicitly required by the event contract.

Continuous engine/wind synthesis resumes from current authoritative state.

17. Deterministic Overflow and Degradation

Audio capacity exhaustion cannot corrupt state or change gameplay.

Stable degradation order is:

1. Drop/defer decorative ambience.
2. Drop ordinary weapon/impact PCM.
3. Reduce optional music voices.
4. Queue or abbreviate ordinary speech.
5. Replace unavailable PCM with SID/text fallback.
6. Duck or temporarily suspend music if necessary.
7. Preserve P1 safety service.
8. Preserve P0 critical warning service.

Every dropped or preempted class has diagnostics suitable for R0/Phase-1 evidence.

18. Implementation Model

AudioEngine is primarily implemented in C.

C owns:

- event arbitration;
- priority scheduler;
- logical voices;
- music state;
- parameter smoothing;
- asset lookup;
- PCM allocation;
- residency checks;
- fallback selection; and
- diagnostics.

Platform wrappers own:

- SID register access;
- PCM/audio-DMA programming;
- protected audio register state;
- channel routing;
- start/stop/retrigger primitives; and
- hardware timing details.

Handwritten 45GS02 assembly is admitted only where required by platform access or supported by measured performance evidence.

19. R0-D / R0-E / R0-F Integration

This white paper does not retroactively change completed R0-A or R0-B evidence and does not require an R0-C retrofit.

New or materially expanded audio contracts are integrated downstream.

Gate	Audio-white-paper disposition
R0-A	No change
R0-B	No retrofit; retain existing hardware/audio evidence for the scope actually tested
R0-C	Finish current work; no audio architecture retrofit
R0-D	Add measurement hooks, representative workloads, counters and protected-latency fixtures
R0-E	Primary combined integration of final scheduler behavior, music/SFX contention, resource staging and simultaneous graphics/input/DMA load
R0-F	Confirm identified R0-E configuration on physical MEGA65

The measured-limits revision freezes only values demonstrated by the corresponding evidence.

20. R0 / Phase-1 Audio Acceptance

The combined audio fixture shall exercise, concurrently where legal:

- continuous dual-engine procedural sound;
- wind;
- three-part minimum music;
- optional additional music voices;

- RWR search/track traffic;
- P0 launch/active-seeker warning;
- master caution;
- P2 critical RIO speech;
- P3 ordinary speech;
- weapon transient PCM;
- PCM preemption;
- missing-sample fallback;
- Attic-to-chip staging;
- renderer DMA load;
- input activity;
- HUD/protected presentation;
- PAL and NTSC supported modes.

Evidence records at minimum:

- event_to_audio_latency
- P0 worst latency
- audio_service_cycles
- PCM channels high-water
- SID voices high-water
- music voices stolen
- PCM preemptions by priority
- dropped effects by priority
- speech interruptions
- cache high-water
- staging bytes
- staging latency
- DMA contention
- audio underrun / distortion faults

No average measurement may substitute for legal worst-phase evidence.

21. Human Audio Acceptance

Technical correctness alone does not establish audio quality.

Human product/audio review must evaluate at minimum:

- engine responsiveness;
- sense of speed;
- stall/buffet communication;
- warning recognizability;
- RWR differentiation;
- own-heater seeker readability;
- speech intelligibility;
- weapon impact;
- carrier launch/recovery impact;
- music identity;
- music/SFX balance;
- warning audibility over music;
- repeated-play fatigue; and
- overall fit with the intended 1985–1992-inspired combat-flight-simulator character.

Warning tones must be recognizable without being unnecessarily painful or fatiguing.

Music must remain identifiable under combat voice stealing.

22. Version 1.0 Baseline

The following decisions constitute the Version 1.0 audio engineering baseline:

- Four SIDs / twelve SID voices are the procedural and music foundation.
- SID 1 and SID 2 form the preferred stereo music domain.
- SID 3 forms the continuous aircraft/environment domain.
- SID 4 forms the warning/RWR/procedural-effect domain.
- Three logical music parts, percussion + bass/rhythm + lead, form the irreducible musical identity.
- Six functional soundtrack roles are the minimum product music scope.
- Four PCM channels form a pooled scheduler-controlled resource rather than two permanently assigned pairs.
- PCM is primarily for speech and short high-information transients.
- Mono is the default PCM asset format conceptually; stereo consumes additional channels only when justified.
- P0/P1 safety presentation outranks music, speech and spectacle.
- Weapon/catapult/impact PCM does not receive safety priority merely because it is sonically important.
- Music is preserved whenever practical but may be temporarily interrupted for protected warning service.
- Every spoken callout retains text fallback.
- Every critical warning retains a protected tone and/or visual fallback.
- Heater growl is own-weapon seeker audio; the F-65 has no incoming IR missile-warning tone.
- Audio hardware never directly consumes Attic addresses.
- ResourceManager may perform bounded Attic-to-chip staging; tactical disk streaming remains prohibited.
- PCM format, rate, duration, cache split, aggregate rate and exact warning-latency ceilings remain evidence-gated.
- AudioEngine owns all logical sound arbitration; platform wrappers alone control protected SID/audio-DMA hardware.
- Audio presentation cannot alter authoritative simulation, sensing, AI, weapons, mission state, replay or checksums.
- New white-paper requirements integrate through R0-D/E/F rather than reopening R0-A/B/C.

End of F-65 Megawing Audio, Sound Effects and Music Engineering White Paper v1.0 – FINAL – HUMAN-REVIEWED ENGINEERING BASELINE